On Latent Predictive Residuals and Automated Threshold Calibration for Multivariate Time-Series Anomaly Detection: An Empirical Investigation

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***Abstract*—Unsupervised anomaly detection in multivariate telemetry requires setting operational thresholds automatically via statistical calibration, such as Extreme Value Theory (EVT/SPOT), without ground-truth labels. While reconstructive baselines like TimesNet and TranAD achieve competitive ranking (PR-AUC 0.29–0.34), heavy-tailed observation residuals cause severe false-alarm flooding, yielding nominal false-positive rates of 24–30%. We investigate whether forecasting in latent representation space via Joint-Embedding Predictive Architectures (JEPAs) conditions residuals for automated thresholding. Across 33 continuous telemetry streams from seven physical domains evaluated under held-out calibration (up to five seeds on the multi-seed subset), unconstrained latent prediction (TS-JEPA) suppresses nominal false-positive rates roughly four-fold down to 7.6% (paired difference -0.153 against TimesNet, $p = 0.002$). However, overall Point-F1 (0.234 vs. 0.197, $p = 0.54$) and PR-AUC (0.323 vs. 0.329, $p = 0.45$) remain statistically indistinguishable from baselines, while detection latency increases relative to the fastest reconstructive baseline ($\Delta t = 80$ vs. 48 steps) and causal event recall drops (10.7% vs. 32.9%). Furthermore, evaluating three candidate physics-inspired regularizers (Koopman spectral entropy, Reynolds stress, and conservative potential flow) reveals domain-specific trade-offs rather than universal gains: linear Koopman operators excel on spacecraft telemetry (PR-AUC 0.591 on SMAP), but curl-free priors fail on cyclic kinematics, and on aggregate, physical regularizers do not outperform unconstrained latent prediction ($p \geq 0.61$). Finally, EVT calibration systematically misses its nominal design risk ($q = 10^{-3}$) on non-stationary telemetry, performing comparably to empirical percentiles. Latent prediction thus provides a more conservative detector that suppresses false alarms, but does not improve ranking fidelity or resolve tail calibration drift.**

***Index Terms*—Multivariate Time-Series, Anomaly Detection, Joint-Embedding Predictive Architecture (JEPA), Residual Conditioning, Threshold Calibration, Extreme Value Theory, Empirical Benchmark.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In remote sensing deployments — NASA SMAP, MSL rovers, industrial IoT facilities, wearable biometric arrays — dozens to hundreds of channels concurrently track power buses, thermal gradients, vibrations, and kinematics [3], and faults must be identified before hardware damage occurs [4].

A. Challenges in Modeling Multi-Sensor Telemetry

Multi-sensor telemetry reflects a physical state trajectory $\dot{x}(t) = \mathbf{f}(x(t))$, yet modeling dynamics in raw observation space faces four obstacles: estimating derivatives $\dot{x} \approx \Delta x / \Delta t$ on noisy discrete streams amplifies high-frequency noise; telemetry captures only a noisy projection of unmeasured states, violating full-state observability; in large arrays ($K \geq 25$) cross-channel nonlinear interactions resist rigid functional forms; and discrete operational mode transitions (orbital day/night cycling, thruster firings, gait freezing) induce continuous distribution drift.

B. Limitations of Deep Anomaly Detectors

Deep detectors face three limitations. First, generative models (autoencoders, VAEs, masked Transformers) minimize pointwise L_1/L_2 errors in raw signal space [3], [5], [6], so reconstruction loss allocates capacity to high-frequency jitter rather than nominal dynamics. Second, contrastive frameworks such as NCAD [7] train on synthetic perturbations whose geometry distorts the embedding space, inducing artificial decision boundaries and false alarms. Third, reconstruction errors $\|x - \hat{x}\|_2^2$ accumulate residual variance across dozens of irrelevant channels; under streaming thresholding (SPOT), the resulting heavy-tailed residuals defeat extreme-value calibrators and trigger false-alarm flooding.

C. Investigating Latent Prediction and Inductive Priors

We study whether latent-space prediction mitigates observation-space noise drowning without synthetic contrastive perturbations. A JEPA maps context x_{ctx} into a representation space and predicts target representations z_{tgt} directly [8], [9], discarding unpredictable stochastic jitter and potentially conditioning residuals for tail calibration. Alongside an unconstrained predictor (`TS-JEPA`), we formalize and test three physics-inspired regularizers: *Operator-Entropy JEPA* (spectral-entropy regularization of a latent Koopman transition

matrix), *Reynolds-Stress JEPA* (second-moment fluctuation-covariance alignment), and *Potential-Flow JEPA* (curl-free gradient dynamics $\hat{v} = -\nabla\Phi$). All share a coordinate-saliency scoring gate and a dyadic frequency front-end under SPOT calibration; a strictly causal input-channel gating variant is evaluated separately in a fault-type study.

D. Contributions

This work makes four principal contributions:

1) Quantifying the Latent Residual Conditioning Effect: Forecasting in abstract representation space rather than reconstructing raw waveforms produces lighter-tailed residuals, lowering held-out nominal FPR from 30.0–30.7% to 7.6% (paired -0.153 vs. TimesNet, p=0.002; -0.151 vs. TranAD, p=0.003). Point-F1/PR-AUC remain statistically indistinguishable, detection latency rises (Δt=80 vs. 48 steps), and causal event recall drops (10.7% vs. 32.9%): latent prediction is a conservative low-false-alarm operating point, not universal dominance.

2) Rigorous Evaluation of Physics-Inspired Inductive Priors: Across 33 streams, neither Operator-Entropy (+0.002 Point-F1, p=0.71) nor Reynolds-Stress (-0.015, p=0.61) beats unconstrained TS-JEPA, and Potential-Flow is significantly worse (-0.081, p=0.033): Koopman operators excel on quasi-linear spacecraft telemetry (SMAP PR-AUC 0.591) but curl-free flows fail on cyclic kinematics. Fixed-backbone ablations show no significant change when the physical terms are disabled ($p \geq 0.375$).

3) Empirical Analysis of Threshold Calibration Failures: EVT/SPOT systematically misses its design risk (q=10^{-3}) on non-stationary telemetry — nominal FPR 5–8% for JEPAs, $\approx$ 22% for baselines — and kurtosis adjustments degrade rather than correct out-of-sample risk control, performing comparably to simple empirical quantiles.

4) Reproducible Multi-Seed Benchmark: 33 matched streams across seven datasets under a strict held-out calibration protocol (65/35 split with isolation gap, up to five seeds, bootstrap CIs, Wilcoxon tests); all code, checkpoints, and logs released.

Section II reviews related work; Section III the benchmark suite; Section IV the formulations; Sections V–VII the protocol, ablations, and results; Section VIII concludes. Supplementary material contains proofs, extended diagnostics, and case-study detail.

II. Related Work

A. Dynamical Systems, Reconstruction, and Forecasting Detectors

Physical modeling of multi-sensor telemetry frames system behavior as continuous trajectories $\dot{x}(t) = \mathbf{f}(x(t))$ [10]. Neural ODEs [11] parameterize velocity fields in raw observation space, where integrating through high-frequency sensor noise induces numerical stiffness and jitter sensitivity. Koopman theory and DMD [12]–[14] instead learn linear operators on observable embeddings; physical disruptions then manifest as spectral decoherence. Our Operator-Entropy variant transfers this idea to a latent transition operator regularized by spectral entropy during training.

The predominant detection paradigm learns nominal patterns via raw-signal reconstruction: LSTM prediction with dynamic thresholding for NASA telemetry [3], association-discrepancy attention in Anomaly Transformer [5], 2D FFT-periodicity decomposition in TimesNet [6], and adversarial sub-series attention in TranAD [15]. These share a vulnerability: point-level losses bias networks toward fitting stochastic noise, and sustained anomalies contaminate the context window. Forecasting alternatives such as DeepAR [16] avoid bidirectional memorization but accumulate variance across irrelevant channels in high-dimensional arrays.

B. Threshold Calibration, Contrastive, and Self-Supervised Detection

Unsupervised detectors require an automated rule converting residual scores to binary alarms without test labels. Non-parametric dynamic thresholding [3] and Streaming Peaks-Over-Threshold (SPOT) [17]—fitting a Generalized Pareto tail above a high quantile under design risk q—are standard; however, EVT's asymptotic guarantees assume i.i.d. tail samples, which telemetry autocorrelation and operational drift violate.

To avoid reconstructing noise, contrastive methods learn discriminative representations: NCAD [7] trains TCNs against injected synthetic anomalies (which distort the physical manifold and miss subtle real faults), and DCdetector [18] maximizes cross-scale consistency via dual-attention contrasting. Joint-Embedding Predictive Architectures [8], [9], [19] instead predict target representations in latent space, with non-contrastive collapse prevention [20]–[22]. Time-series extensions [23]–[25] typically evaluate ranking metrics or post-hoc threshold sweeps without studying how latent residuals behave under automated tail calibration—the gap this work addresses.

C. Physics-Inspired Inductive Biases

PINNs [26] and Hamiltonian/Lagrangian networks [27], [28] embed dynamics as hard constraints; their stiff competing gradients [29] and closed-form-PDE requirements fit poorly to uncalibrated multi-sensor telemetry. We instead test whether *inspired* structures—spectral entropy, second-moment closure, conservative potential flow—improve latent residuals.

D. Clarification of Technical Terminology

- **Physics-inspired vs. physics-informed:** the latter enforces known governing equations as loss residuals; the former adopts physics-motivated structures as exploratory regularizers, without claiming latent coordinates are physical state variables.
- **Causal preprocessing vs. causal decision-time access:** the former computes features strictly from history ($\partial \mathbf{g}/\partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$); the latter specifies whether alarms are issued with 0-lookahead or after an S-step buffer.
- **Conservative vs. dissipative:** autonomous gradient flows $\dot{z} = -\nabla\Phi$ satisfy $\frac{d}{d\tau}\Phi \leq 0$ and preclude periodic orbits.
- **Operator non-degeneracy vs. representation anti-collapse:** the transition operator $\mathbf{K}$ not collapsing to zero is distinct from the encoder not collapsing representations to a constant.

E. Positioning

This investigation bridges joint-embedding predictive learning and multivariate anomaly detection: forecasting latent z_{tgt} rather than reconstructing $\hat{x}$ avoids noise drowning; three candidate physics-inspired regularizers are evaluated empirically against unconstrained latent controls; latent coordinate-saliency gating and dyadic frequency shells preprocess residuals; and EVT/SPOT calibration is evaluated on strictly held-out nominal telemetry to quantify the gap between design risk and empirical false-alarm rates.

III. Multi-Sensor IoT and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams sourced from NASA spacecraft archives [3], the UCI Machine Learning Repository [30], [31], TimeEval [32], and mTSBench [33]: 33 streams are evaluated under the primary multi-seed held-out calibration protocol and 12 reserved for operational case studies. Streams span four physical domains: *spacecraft telemetry* (SMAP Earth-observation `A-1`/`A-2`/`A-7`/`D-3`/`P-3`/`E-1` power, thermal, and attitude subsystems; MSL rover `C-1`/`C-2`/`D-14`/`D-15`/`D-16`/`M-1` actuator and environmental channels, with two short-nominal streams reserved; SWAN-SF solar magnetograms [34]); *wearable kinematics* (six Daphnet FoG streams [30]: 64 Hz tri-axial accelerometry for Freezing-of-Gait events, so S=64 is a 1-s horizon); *industrial processes* (14 GHL gasoil heating-loop streams, the Genesis pick-and-place testbed, and GECCO water-distribution chemistry/hydrology at 1-minute resolution); and *smart infrastructure* (Room-Occupancy environmental sensing [35], six CICIDS edge network-flow streams, CalIt2 people-flow [31], and Metro I-94 highway traffic).

Each stream provides nominal training sequences and labeled anomaly test sequences; channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatline phases. We evaluate the standardized subsets studied in the literature [3], [32] (of 55 SMAP and 27 MSL archive entities) for strict matching and reproducible runtime.

To eliminate circular in-sample calibration, each stream's nominal sequence is partitioned chronologically into training (65%), calibration (17.5%), and held-out evaluation (17.5%) blocks separated by an isolation gap (Section V-A). The held-out protocol spans 33 streams across seven datasets (6 Daphnet, 14 GHL, 4 MSL, 6 SMAP, Genesis, Metro, SWAN); streams whose nominal blocks are too short for disjoint partitions (CalIt2, Room-Occupancy, two MSL) or where multi-seed stride-1 evaluation over 10^6 steps was prohibitive (GECCO, CICIDS) are reserved as case studies.

Supplementary Table **??** specifies the suite: 2,079,507 nominal training and 2,986,574 test timesteps (>5.06M observations), with per-dataset domain, channel dimension K, sequence lengths, contamination ratio $\pi\%$, anomaly typologies, and partition protocols.

A machine-readable manifest (`reports/stream_manifest.csv`, generated by `scripts/build_stream_manifest.py`) enumerates every stream's identifier, channel dimension, split lengths, held-out anomaly counts, benchmark role, and evaluated seed set as recorded in the released artifacts. All counts in this section, the abstract, and the results tables reconcile against it: 33 primary held-out streams (19 non-GHL at five seeds, 14 GHL at three to five seeds depending on size) and 12 case-study streams.

IV. Methodology: Latent Representation Forecasting and Structured Inductive Priors

This section formalizes the mathematical formulation, latent predictors, and inductive biases of the proposed framework.

A. Problem Formulation and Latent Space Prediction

Let $\mathbf{X} \in \mathbb{R}^{T\times K}$ denote a multivariate stream partitioned into context $x_{\text{ctx}} = \mathbf{X}[t-C:t,:] \in \mathbb{R}^{C\times K}$ and suspect horizon $x_{\text{tgt}} = \mathbf{X}[t:t+S,:] \in \mathbb{R}^{S\times K}$. Reconstructive baselines (TimesNet, TranAD, Anomaly Transformer) minimize pointwise error $\min_\theta \mathbb{E}\|\mathbf{X} - \mathcal{D}_\theta(\mathcal{E}_\theta(\mathbf{X}))\|_F^2$, where ambient jitter dominates the Frobenius norm and capacity is spent reconstructing noise (noise drowning). We instead adopt the JEPA framework [8]: a context encoder E_θ maps $x_{\text{ctx}} \to z_{\text{ctx}} \in \mathbb{R}^D$ and an EMA momentum target encoder E_ϕ maps $x_{\text{tgt}} \to z_{\text{tgt}}$, with $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$, $m_k \in [0.996, 0.9995]$. The objective forecasts target representations in latent space, testing whether structured priors condition prediction residuals for tail calibration.

B. Latent Predictor Architectures and Dynamical Inductive Priors

To examine whether incorporating structured dynamical inductive biases stabilizes representation forecasting, we evaluate an unconstrained latent predictor alongside three candidate formulations inspired by physical dynamical principles:

1) Operator-Entropy JEPA (Spectral Dynamics & Modal Dispersion): Motivated by Koopman operator theory [36], [37], in which nonlinear trajectories are analyzed via linear operators on observable functions, we parameterize the predictive latent transition via a learned operator $\mathbf{K} \in \mathbb{R}^{D\times D}$: $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$. To prevent collapse onto a low-rank subspace (dimensional collapse) or to zero (energy collapse), the regularizer decouples spectral uniformity from total operator energy.

Let $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ denote the singular values of $\mathbf{K}$, $\lambda_i = \sigma_i^2$ the eigenvalues of $\mathbf{K}\mathbf{K}^T$, and $\mathcal{E}(\mathbf{K}) = \|\mathbf{K}\|_F^2 = \sum_i \lambda_i$ the total energy. For $\mathcal{E}(\mathbf{K}) > 0$, the scale-free spectral distribution over modes is $p_i = \sigma_i^2/\|\mathbf{K}\|_F^2$, $\sum_i p_i = 1$. We omit isotropic diagonal loading from the Gram matrix before normalization — it would give $\mathbf{K} \to \mathbf{0}$ an artificial identity spectrum with maximal entropy, spuriously rewarding collapse. The Von Neumann spectral entropy uses the convention $0\log 0 = 0$:

$$\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i:p_i>0} p_i \log p_i. \tag{1}$$

The *Spectral Entropy Deficiency* quantifies the Kullback-Leibler divergence from the uniform modal equipartition $\mathbf{u} = [1/D, \ldots, 1/D]^T$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \,\|\, \mathbf{u}) \geq 0. \tag{2}$$

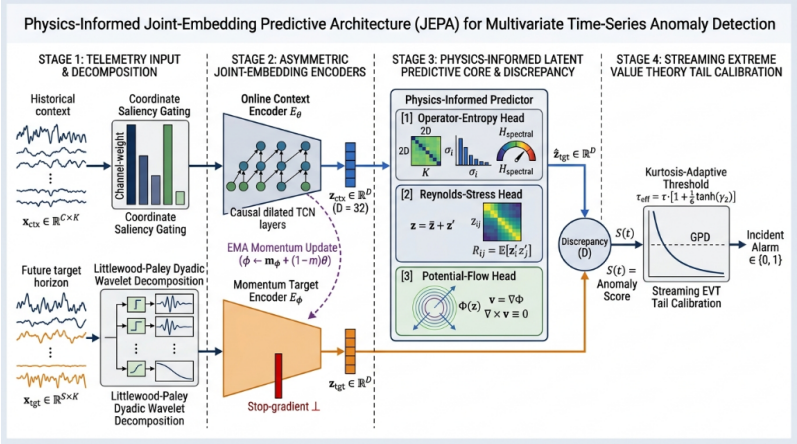


Fig. 1. JEPA telemetry-monitoring architecture: context and suspect windows are encoded into z_{ctx}, z_{tgt} and compared in latent space (avoiding observation-space false-alarm flooding), under three candidate physical regularizers (Koopman spectral entropy, covariance closure, potential flow) with held-out SPOT calibration.

Here, $\Delta\mathcal{H} = 0$ if and only if all modes carry equal energy ($\sigma_1 = \cdots = \sigma_D$), while dimensional collapse to rank 1 yields maximal deficiency $\Delta\mathcal{H} = \log D$.

To penalize energy collapse ($\|\mathbf{K}\|_F \to 0$) and prevent explosive gain, we introduce a non-negative Bregman energy barrier relative to nominal isometric scale $\mathcal{E}_0 = \gamma D$ ($\gamma = 1.0$):

$$\Phi_{\text{energy}}(\mathbf{K}) = r_{\text{safe}}(\mathbf{K}) - 1 - \log r_{\text{safe}}(\mathbf{K}) \geq 0, \quad r_{\text{safe}}(\mathbf{K}) = \max\left(\frac{\|\mathbf{K}\|_F^2}{\gamma D}, \epsilon\right) \tag{3}$$

where $\epsilon = 10^{-6}$ provides a numerical floor. For the identity operator $\mathbf{K} = \mathbf{I}$, $\|\mathbf{I}\|_F^2 = D$, yielding $\Phi_{\text{energy}}(\mathbf{I}) = 0$ exactly. We emphasize that the implemented barrier is *floored*: it attains a finite maximum penalty $\epsilon - 1 - \log\epsilon \approx 12.8$ (before β_{norm} weighting) throughout $\|\mathbf{K}\|_F^2 < \epsilon\gamma D$, rather than diverging; divergence to $+\infty$ is a property only of the unfloored idealization $\Phi^*_{\text{energy}} = r - 1 - \log r$. The operator is initialized near identity ($\mathbf{K} \sim \mathbf{I} + \mathcal{N}(\mathbf{0}, 0.01^2\mathbf{I})$), so $r_{\text{safe}} \approx 1$ and training starts with a vanishing energy penalty; since LayerNorm gives $\mathbb{E}[\|z\|_2^2] \approx D$, the isometric scale $\mathcal{E}_0 = \gamma D$ matches the natural energy of unitary transitions. In sensitivity checks across $\lambda_{\text{ent}} \in \{0.1, 0.5, 1.0\}$ the model converged stably, with 0.5 balancing spectral spread against predictive regression. The composite spectral regularizer is:

$$\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K}), \tag{4}$$

with $\beta_{\text{norm}} = 1.0$. The total training objective is:

$$\mathcal{L}_{\text{Op-Entropy}} = \|\mathbf{K}z_{\text{ctx}} - z_{\text{tgt}}\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K}), \tag{5}$$

with $\lambda_{\text{ent}} = 0.5$ in all experiments. All shared representation components (saliency gating, dyadic frequency shells, LayerNorm) are held fixed across models and controls, isolating the transition-operator loss. We emphasize that $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ acts on the operator matrix, enforcing a non-collapsed operator spectrum; it provides no analytical guarantee against encoder representation collapse, which is mitigated empirically by the EMA target encoder and stop-gradient operations [9], [38].

Proposition 1 (Operator Triviality Invariance)**.** *Let* $\mathcal{L}(\mathbf{K}) = \frac{1}{B}\sum_{b=1}^{B}\|\mathbf{K}z_{ctx}^{(b)} - z_{tgt}^{(b)}\|_2^2 + \lambda_{ent}\mathcal{R}_{spectral}(\mathbf{K})$ *with* $\lambda_{ent} > 0, \beta_{norm} > 0$*, non-trivial target batch* $\frac{1}{B}\sum_{b=1}^{B}\|z_{tgt}^{(b)}\|_2^2 = E_{tgt} > 0$*, and non-degenerate context–target coupling* $\frac{1}{B}\sum_{b=1}^{B} z_{tgt}^{(b)}\, z_{ctx}^{(b)^T} \neq \mathbf{0}$*. Then (i) the zero operator* $\mathbf{K} = \mathbf{0}$ *is strictly not a local or global minimizer of* $\mathcal{L}(\mathbf{K})$ *under the implemented floored barrier, and (ii) under the unfloored idealization* ($\epsilon \to 0^+$, $\Phi^*_{energy} = r - 1 - \log r$)*, additionally* $\lim_{\|\mathbf{K}\|_F \to 0}\mathcal{L}(\mathbf{K}) = +\infty$.

Proof sketch. At $\mathbf{K} = \mathbf{0}$ the floored barrier is saturated and contributes zero gradient, while $\nabla_{\mathbf{K}}\mathcal{L}_{\text{pred}}|_{\mathbf{0}} = -\frac{2}{B}\sum_b z_{\text{tgt}}^{(b)}\, z_{\text{ctx}}^{(b)^T} \neq \mathbf{0}$, so $\mathbf{K} = \mathbf{0}$ is non-stationary; (ii) follows from $-\log r \to +\infty$ while $\mathcal{L}_{\text{pred}} \to E_{\text{tgt}} < \infty$. Full proof in Supplementary Section **??**. □

Remark (Operator Non-Degeneracy vs. Representation Collapse): Proposition 1 concerns only the transition operator: it does not prevent encoder collapse. If E_θ mapped all inputs to a constant $z^{(b)} = \mathbf{c}$, then $\mathbf{K} = \mathbf{I}$ would attain zero prediction error and zero spectral penalty while encoding no information; prevention of that failure mode rests empirically on the stop-gradient momentum dynamics, not on Proposition 1.

2) Reynolds-Stress JEPA (Second-Moment Closure Head): Reynolds decomposition splits a turbulent velocity field into a mean flow and a fluctuation component, where the Reynolds stress tensor captures the second moment of velocity fluctuations. We adapt this second-moment formulation as a latent-space inductive bias; the latent coordinates are learned embeddings, and $\mathbf{\Sigma}$ is not a physical momentum-transport tensor.

A two-layer MLP predicts the conditional mean target latent $\hat{z}_{\text{tgt}}$ from z_{ctx}, and a learned linear projection $\mathbf{P} \in \mathbb{R}^{d_s \times D}$ ($d_s = 16$) applied to the EMA target representation yields the empirical second-moment matrix:

$$\mathbf{\Sigma}_{\text{obs}} = (\mathbf{P}z_{\text{tgt}})(\mathbf{P}z_{\text{tgt}})^T + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s \times d_s}. \tag{6}$$

Because $\mathbf{\Sigma}_{\text{obs}}$ is computed from an instantaneous latent vector rather than averaged over the target horizon S, it forms a rank-one outer product with a ridge term enforcing membership in $\mathbb{S}_{++}^{d_s \times d_s}$; statistically it is an auxiliary second-moment projection penalty. The closure head $h_\psi(z_{\text{ctx}})$ outputs a lower-triangular $\mathbf{L} \in \mathbb{R}^{d_s \times d_s}$ with positive diagonal $L_{ii} = \text{softplus}(a_{ii}) + \epsilon$, and the predicted covariance is reconstructed via Cholesky factorization:

$$\mathbf{\Sigma}_{\text{pred}} = \mathbf{L}\mathbf{L}^T + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s \times d_s}, \tag{7}$$

confining predictions to $\mathbb{S}_{++}^{d_s \times d_s}$. The training objective is:

$$\mathcal{L}_{\text{Reynolds}} = \text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}}) + 0.1 \text{ MSE}(\mathbf{\Sigma}_{\text{pred}}, \mathbf{\Sigma}_{\text{obs}}). \quad (8)$$

At inference time, the anomaly score combines the mean latent discrepancy and the fluctuation tensor residual via the coordinate-saliency gate (formalized in Section IV-C). The stress term enters both the training loss and the test score; its downstream contribution is evaluated in Section VII-B and Table I(a).

3) Potential-Flow JEPA (Conservative Field Theory and Flow-Matching Derivation): Under Helmholtz-Hodge decomposition, a smooth field $v \in C^1(\mathbb{R}^D, \mathbb{R}^D)$ splits into a curl-free potential gradient and a solenoidal field, $v = -\nabla\Phi + \nabla \times A$. As dissipative systems relax toward local minima on potential-energy surfaces, we model latent evolution as the dissipative gradient flow $\dot{z}(\tau) = -\nabla_z \Phi_\psi(z(\tau))$ for a scalar potential network Φ_ψ with smooth (SiLU) activations.

Proposition 2 (Absence of Periodic Attractors in Gradient Flows)**.** *Let $\dot{z} = -\nabla\Phi(z)$ be an autonomous gradient flow on $\mathbb{R}^D$ where $\Phi \in C^2(\mathbb{R}^D, \mathbb{R})$. There exist no non-constant periodic orbits.*

Proof sketch. Along a closed orbit, $0 = \Phi(\gamma(T_p)) - \Phi(\gamma(0)) = -\int_0^{T_p} \|\nabla\Phi(\gamma(t))\|_2^2 dt$, forcing $\nabla\Phi \equiv \mathbf{0}$ and hence $\dot{\gamma} \equiv \mathbf{0}$, contradicting non-constant periodicity. Full proof in Supplementary Section **??**. □

a) Continuous Velocity Field and Multi-Subspace Regime Routing: The velocity field is the negative gradient of a parameterized scalar potential, $\hat{v}(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) = -\nabla_{z_t} \Phi_\psi(z_t, t, z_{\text{ctx}}, p_{\text{regime}})$, where Φ_ψ is a three-layer residual MLP (LayerNorm, SiLU) conditioned on state z_t, diffusion time $t \in [0, 1]$ (sinusoidal embedding), context z_{ctx}, and a discrete regime projection p_{regime}; the raw gradient is tangent-projected via $v_\perp = v - \langle u, v\rangle u$, $u = z_t/\|z_t\|_2$, bounding integration curvature.

Discrete operational modes use $K_r = 4$ learnable orthonormal subspace projections $\mathbf{U}_k \in \mathbb{R}^{D \times d}$ (d=8, D=32); the active regime is selected by maximal captured context energy, $k^* = \arg\max_k \|\mathbf{U}_k^\top z_{\text{ctx}}\|_2^2$, $p_{\text{regime}} = \mathbf{U}_{k^*}\mathbf{U}_{k^*}^\top z_{\text{ctx}}$, with an orthogonality separation loss $\mathcal{L}_{\text{subspace}} = \frac{2}{K_r(K_r-1)} \sum_{i<j} \|\mathbf{U}_i^\top \mathbf{U}_j\|_F^2$. Section VI-A includes a single-regime control (K_r=1) with routing disabled.

b) Flow-Matching Objective, Numerical Midpoint Evaluation, and Dynamical Misspecification: Proposition 2 governs the autonomous continuous flow $\dot{z} = -\nabla\Phi(z)$; the empirical objective instead trains $\hat{v}$ via conditional flow matching (OT-CFM) along linear paths between noise $z_{\text{noise}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and z_{tgt}: $z_t = (1-t)z_{\text{noise}} + t z_{\text{tgt}}$, $t \sim \mathcal{U}(0, 1)$, with target field $v_t^* = z_{\text{tgt}} - z_{\text{noise}}$ and loss

$$\mathcal{L}_{\text{Pot-Flow}} = \mathbb{E}_{t, z_{\text{noise}}} \left[\|\hat{v}(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) - (z_{\text{tgt}} - z_{\text{noise}})\|_2^2 \right]. \quad (9)$$

Along the deterministic mean trajectory ($z_{\text{noise}} = \mathbf{0}$), $z_t = t\, z_{\text{tgt}}$ and $v_t^* = z_{\text{tgt}}$ is constant.

At inference, rather than integrating an ODE over $t \in [0, 1]$, the detector queries the field at the trajectory midpoint:

$$\mathbf{e}(t) = \hat{v}(0.5 z_{\text{tgt}}, 0.5, z_{\text{ctx}}, p_{\text{regime}}) - z_{\text{tgt}}. \quad (10)$$

Since the reference velocity along the mean path is $v_{0.5}^* = z_{\text{tgt}}$, this midpoint query directly measures the discrepancy between the learned potential gradient and the nominal target, matching the dimensionality (D=32) and scale of the unconstrained TS-JEPA residual. The anomaly score applies the saliency gate: $\mathcal{S}(t) = g(\mathbf{e}(t))$.

Dynamical Misspecification on Non-Conservative Telemetry: A curl-free velocity field ($\nabla \times \hat{v} \equiv \mathbf{0}$) forces circulation along any closed path $\mathcal{C}$ to vanish, $\oint_{\mathcal{C}} \hat{v} \cdot d\mathbf{r} = 0$. Valid for purely dissipative equilibria, this is dynamically misspecified for the non-conservative cyclic dynamics common in telemetry (motor revolutions, diurnal GECCO consumption cycles, periodic Daphnet gait kinematics): normal circulating orbits are penalized as anomalous. This predicts degraded detection on cyclic telemetry, tested in Section VII-B and Supplementary Section **??**.

C. Anomaly Scoring and Operational Detection Latency

During inference, context windows $x_{\text{ctx}} = \mathbf{X}[t - C : t, :]$ and suspect windows $x_{\text{tgt}} = \mathbf{X}[t : t + S, :]$ are processed at stride 1. For TS-JEPA, Operator-Entropy, and Potential-Flow, the score gates the latent discrepancy through $g(\cdot)$ (Section IV-D1): $\mathcal{S}(t) = g(\hat{z}_{\text{tgt}}(t) - z_{\text{tgt}}(t))$. For Reynolds-Stress, $\mathcal{S}(t) = g(z_{\text{tgt}} - \hat{z}_{\text{tgt}}) + 0.5\, g(\text{vec}(\mathbf{\Sigma}_{\text{obs}} - \mathbf{\Sigma}_{\text{pred}}))$. All shared representation stages are held fixed across variants and ablation controls (Section VI-A).

Operational Distinction: Zero-Lookahead Causal Filtering vs. Buffered Predictive Early Warning. Two regimes exist: (1) zero-lookahead filtering, emitting $\hat{y}_t$ from observations up to t; and (2) buffered predictive monitoring across lookahead S. TS-JEPA uses regime (2): given context x_{ctx}, it projects $\hat{z}_{\text{tgt}}$ over $[t, t+S]$, so an anomaly at t^* can raise discrepancy up to S=64 steps before onset. Under strict unadjusted point-level evaluation (stride 1), warnings during $[t^* - S, t^*]$ score as false alarms — predictive foresight incurs a retrospective metric penalty for early warning. Incident analysis (Supplementary Section **??**) also evaluates *Middle* alignment ($t + S/2$); all baselines share identical windowing, horizon, and standardization.

D. Shared Internal Inductive Biases

To stabilize learning across high-dimensional, noisy telemetry, all three architectures incorporate three inductive mechanisms:

1) Coordinate Saliency Gating and the Auxiliary Channel-Gate Variant: On high-dimensional streams ($K \geq 25$), faults frequently originate in a localized subsystem while the remaining channels exhibit ambient noise, diluting localized anomalies in isotropic representations. The deployed detector mitigates this in latent space via an anti-drowning coordinate saliency gate on error vectors $\mathbf{e} = z_{\text{tgt}} - z_{\text{pred}} \in \mathbb{R}^D$:

$$\mathcal{D}_{\text{sal}}(\mathbf{e}) = (1-\alpha)\|\mathbf{e}\|_2 + \alpha\sqrt{D \sum_{d=1}^{D} w_d e_d^2}, \quad w_d = \frac{\exp(|e_d|/\tau)}{\sum_{j=1}^{D} \exp(|e_j|/\tau)}, \quad (11)$$

amplifying sparse coordinate deviations by up to $\sqrt{D}$ while remaining exactly $\|\mathbf{e}\|_2$ under isotropic errors. This is the discrepancy gate used in every experiment: input channels

are encoded unweighted, and the gate acts only on the D-dimensional predictive residual.

a) Auxiliary Input-Channel Gate (Evaluated Variant): We also formulate a strictly causal *Causal Channel Saliency Gate* as an auxiliary variant (full specification and the fault-type study in Supplementary Section **??**): each channel is weighted by the sigmoid of its standardized context-window variance share, $g_k = \sigma((s_k - \bar{s})/(\sigma_s + \epsilon))$, applied symmetrically to context and target encoders so that $\partial g_k / \partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$ (zero target leakage). For $K{=}1$ the implementation short-circuits to identity $\mathbf{g} = \mathbf{1}$ (the naive formula degenerates to 0.5); equal variance shares yield the neutral value 0.5 — a *relative* reweighting absorbed by downstream LayerNorm, not a normalized partition — and a constant channel is suppressed toward 0, not removed. Boundary cases verified numerically: $K{=}1 \Rightarrow g \equiv 1.0$; i.i.d. eight-channel input gives mean 0.50 ± 0.22; a $10\times$-variance channel gates to 0.93 vs. 0.41 for the rest; a constant channel to 0.08; and an abrupt mid-context variance shift gates identically to a persistent one (0.93), since shares pool over the context window.

b) Zero-Leakage Data Provenance and Temporal Isolation: We enforce strict temporal isolation across the training, nominal-calibration, and held-out evaluation partitions (Section V-A): standardization statistics come only from the training partition, EVT tail parameters (u, σ, ξ) only from the calibration partition, and evaluation proceeds chronologically with frozen weights and thresholds. The auxiliary gate's weights depend only on x_{ctx}; the deployed pipeline weights inputs uniformly.

c) Gating Boundary Conditions and Failure Modes: Three boundary conditions apply to variance-based gating. First, intrinsically quiet channels keep low s_k and remain attenuated even during small absolute-magnitude anomalies; streams are therefore standardized by nominal statistics before gating. Second, quiet faults (flatlines, stuck values, $v_k^{\text{ctx}} \to 0$) cannot be isolated by variance amplification; the fault-type study (Section **??**) confirms they are weakly detected under every gating regime (recall 0.02–0.07), a limitation of the latent discrepancy signal rather than of gating — flatlines are instead caught, only partially, through predictive discrepancy when the faulted channel contributes to the aggregate residual. Third, abrupt regime transitions transiently perturb variance shares until the $C{=}256$ context window refills.

2) Littlewood-Paley Dyadic Wavelet Shells: Anomalies span temporal scales from sensor spikes to thermal drifts; the encoder front-end therefore decomposes inputs into $J{=}4$ dyadic frequency bands. Continuous wavelet theory defines the Littlewood–Paley partition of unity as:

$$|\phi(\omega)|^2 + \sum_{j=0}^{J-1} |\psi(2^j \omega)|^2 = 1, \quad \forall \omega \in [-\pi, \pi], \tag{12}$$

discrete numerical implementations require exact partition enforcement on a finite frequency grid. Over an $N_{\text{fft}} = 256$ discrete real fast Fourier transform (rFFT) grid yielding $F = 129$ non-negative frequency bins $\omega_m = m/N_{\text{fft}}$ for $m \in \{0, \ldots, N_{\text{fft}}/2\}$, dyadic center frequencies are positioned at $c_j = 2^{j-J+1}$ for $j \in \{0, 1, 2, 3\}$. The unnormalized shells are constructed in three branches: a half-Gaussian profile anchored at DC for $j = 0$, log-frequency octave bells $\tilde{\Psi}_j(\omega_m) = \exp\left(-\frac{(\log_2 \omega_m - \log_2 c_j)^2}{2\sigma^2}\right)$ with bandwidth $\sigma = 0.6$ for interior shells $j \in \{1, 2\}$, and a plateau rising toward the Nyquist edge for $j = J - 1$. To strictly preserve the partition property across all discrete frequencies, filter weights are normalized bin-wise into an exact partition of unity:

$$\Psi_j[\omega_m] = \frac{\tilde{\Psi}_j(\omega_m)}{\sum_{l=0}^{J-1} \tilde{\Psi}_l(\omega_m)}, \quad \sum_{j=0}^{J-1} \Psi_j[\omega_m] \equiv 1.0, \quad \forall m \in \{0, \ldots, F{-}1\}. \tag{13}$$

The frequency-domain split is applied directly via $\mathbf{x}_j = \text{irfft}(\Psi_j \odot \text{rfft}(\mathbf{x}))$, which is numerically exact: measured partition-of-unity deviation is below 1.2×10^{-7} and reconstruction error below 5×10^{-8}. Each shell is processed by a full (non-depthwise) causal 1D convolution with kernels $k \in \{15, 9, 5, 3\}$ (coarse-to-fine) and left-side causal padding $p = k - 1$, then recombined by a learned softmax scale router over pooled per-shell energies, a 1×1 projection, residual scaling $\gamma = 0.1$, and LayerNorm. The normalized shells overlap smoothly at band edges, so band energies are correlated ($\sum_j \|\mathbf{x}_j\|^2$ recovers $\approx 70\%$ of signal energy) and the finest shell concentrates near Nyquist. Comparing each band's spectral mass inside its ideal dyadic octave $[2^{-(j+1)}, 2^{-j}]$ against Mallat-form Haar and Daubechies-4 octave responses (`reports/lp_filterbank_diagnostics.csv`): our shells place $\{0.50, 0.33, 0.20, \approx 1.00\}$ in-band versus $\{0.78, 0.50, 0.55, 0.82\}$ (Haar) and $\{0.86, 0.63, 0.64, 0.87\}$ (DB4) — the front-end is *less* frequency-localized than either standard wavelet bank, trading octave isolation for smooth, exactly-reconstructing amplitude partitions and learned scale routing. We therefore characterize the block as a Fourier-domain multiscale front-end with an exact discrete partition of unity—not as an exact Littlewood-Paley octave decomposition—and rely on the matched on/off ablation in Section VI (Table Ib) to measure its empirical contribution. This decomposition isolates transient high-frequency perturbations from low-frequency diurnal baselines before latent projection.

3) Kurtosis-Adaptive Extreme Value Theory (EVT) and Calibration Realities: Binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_t)$ are generated via Streaming Peaks-Over-Threshold (SPOT) [17]: excesses over a high threshold u (98th percentile) asymptotically follow a Generalized Pareto Distribution (GPD) with scale σ and shape ξ:

$$\tau = u + \frac{\sigma}{\xi}\left[\left(\frac{qN}{N_u}\right)^{-\xi} - 1\right], \tag{14}$$

where $q = 10^{-3}$ denotes the design risk probability, N is total calibration samples, and N_u is the count of excesses. To account for varying tail heaviness across diverse sensor regimes, we modulate the threshold via an empirical heuristic factor governed by the sample excess kurtosis γ_2 of calibration residuals:

$$\tau_{\text{eff}} = \tau \cdot \left[1 + \frac{1}{6}\tanh(\gamma_2)\right]. \tag{15}$$

On heavy-tailed distributions ($\gamma_2 > 0$), τ_{eff} rises to suppress volatile-noise false alarms; on light-tailed ones it tightens for sensitivity.

We emphasize that this is an engineering heuristic for pre-asymptotic finite-sample tails, not an asymptotic EVT theorem. In the controlled calibrator ablation (Section VII-D, Table IV) it does not improve out-of-sample risk control: on held-out data the multiplier raises TS-JEPA nominal FPR from 5.26% to 17.68% with no statistically distinguishable Point-F1 change (0.1694 vs. 0.1577).

More broadly, EVT does not eliminate threshold-calibration difficulty: under $q = 10^{-3}$ (ideal asymptotic FPR 0.1%), measured held-out nominal FPR settles at 5.3–6.2% for the JEPAs and $\approx 22\%$ for TimesNet/TranAD, owing to finite-sample variance, residual autocorrelation, and non-stationary drift between nominal training and deployment. The calibrated finding of this paper is narrower: latent predictive residuals produce substantially fewer false alarms than observation-space residuals under identical calibration, reflecting lighter tails at the cost of lower causal recall. Thresholds use static EVT on held-out nominal splits; dynamic rolling SPOT is evaluated separately in the operational incident analysis.

V. Experimental Protocol and Evaluation Metrics

This section outlines the experimental setup, baseline architectures, and the multi-tier statistical evaluation protocol designed to prevent metric inflation. The master experiment matrix enumerating all evaluation suites (stream counts, train/calibration/test partitions, horizons, calibrators, and metrics) is provided in Supplementary Table **??**.

A. Experimental Setup and Hyperparameters

To ensure rigorous and fair comparison, all architectures are trained and evaluated under an identical streaming time-series pipeline. Telemetry streams are processed using a sliding window with past context duration $C = 256$ and suspect target horizon $S = 64$ (total window length $L = 320$), evaluated with a stride of 1 timestep across all testing streams.

Context representations $z_{\text{ctx}} \in \mathbb{R}^D$ ($D = 32$) are extracted using a causal dilated Temporal Convolutional Network (TCN) comprising 3 residual blocks with channel depth $F = 48$, kernel size $K = 5$, and dilation factors $d \in \{1, 2, 4\}$. While the convolutional receptive field across the 3 dilated blocks spans $1 + 2(K-1)\sum_l d_l = 57$ timesteps, global temporal average pooling across the resulting feature maps aggregates information across the entire context window $C = 256$, condensing the sequence into a fixed-dimensional latent representation $z_{\text{ctx}} \in \mathbb{R}^D$ regularized by Layer Normalization and dropout ($p = 0.20$). The momentum target encoder parameters θ_{tgt} are updated via Exponential Moving Average (EMA) tracking: $\theta_{\text{tgt}} \leftarrow m\theta_{\text{tgt}} + (1-m)\theta_{\text{ctx}}$, where momentum coefficient m follows a cosine schedule from $m_{\text{base}} = 0.996$ to $m_{\text{max}} = 0.9995$.

Models are optimized using AdamW with an initial learning rate $\eta = 10^{-3}$ decayed to $\eta_{\text{min}} = 10^{-5}$ via Cosine Annealing, weight decay 10^{-4}, and mini-batch size $B = 32$ across 20 training epochs on nominal telemetry sequences.

a) Held-Out Calibration Protocol: To evaluate out-of-sample threshold calibration without circularity, each stream's nominal telemetry is partitioned into three disjoint chronological blocks: *training* (first 65%; representation weights only), *nominal calibration* (next 17.5%; withheld from training, used solely to fit per-stream GPD tail parameters σ, ξ above u=0.98 at q=10^{-3}, with γ_2 estimated strictly within this block), and *nominal evaluation* (final 17.5%, separated by an unobserved 64-step isolation gap; held-out FPR measured on data the threshold never saw). Thresholds then remain fixed during test evaluation to measure static EVT stability under drift; dynamic rolling SPOT is evaluated separately in the operational incident analysis (Supplementary Section **??**).

b) Calibrator Assignment Across Experiments: To avoid ambiguity, we specify the calibrator per experiment. **Primary held-out benchmark** (Tables II, III, IV; all paired statistics): standard SPOT ($u = p_{98}$, $q = 10^{-3}$, no kurtosis multiplier); headline metrics are the "SPOT" column of Table IV, with calibrator variants compared on identical score streams. **Ablations and controlled studies** (Section VI): SPOT with the $\tanh(\gamma_2)/6$ kurtosis multiplier (the released `adaptive_kurtosis` flag). **Early-warning and EVT mechanism studies**: standard SPOT unless the calibrator is the object of comparison. **Incident metrics** (Supplementary Section **??**): rolling SPOT. No experiment mixes calibrators across models within a table.

c) Multi-Seed Replication, Primary Endpoints, and Multiplicity Controls: Principal models on the 19 non-GHL streams are trained across five seeds (42, 123, 456, 7, 2024); the 14 large GHL streams ($> 10^6$ steps) use three seeds (five on the two smallest) for cost, and ablation controls use three. Two co-primary endpoints: (i) *nominal FPR reduction* on held-out nominal data, testing whether latent prediction stabilizes EVT bounds; and (ii) *detection parity* via unadjusted Point-F1 and PR-AUC, testing whether suppression degrades discrimination.

Inference reports stream-macro means, seed SDs, moving-block bootstrap 95% CIs (block L=64), and two-sided Wilcoxon signed-rank tests. The independent unit of analysis is the telemetry stream (n=19 non-GHL at five seeds; n=33 where GHL enters at available coverage): seeds are averaged within stream and repeated windows aggregated to one stream-level score, so nothing is double-counted. Because nonsignificant tests do not establish equivalence, we make no noninferiority claim: paired differences are reported with effect sizes and CIs for readers to judge practical margins. p-values across comparisons are Benjamini–Hochberg FDR-adjusted at $q_{\text{FDR}} = 0.05$; CIs spanning zero or adjusted $p > 0.05$ are treated as inconclusive, not as equivalence.

B. Baseline Architectures and Parameter Capacities

To isolate algorithmic principles from external implementation discrepancies, all baselines are executed within a unified PyTorch benchmarking pipeline under identical windowing, standardization, and EVT calibration:

TimesNet [6]: Multi-periodic reconstructive architecture mapping 1D series to 2D variation spaces via FFT period discovery and Inception blocks (648,553 parameters at K=25; 574,529 at K=9).

TranAD [15]: Adversarial reconstructive Transformer with two-stage self-conditioning and sub-series attention (205,746 at K=25; 201,090 at K=9).
TS-JEPA (Unconstrained Control): Standard JEPA predicting z_{tgt} from z_{ctx} via a 2-layer MLP without physical priors (289,061 at K=25; 218,309 at K=9; only the input projection scales with K at $+48$ parameters per channel).
Physics-Inspired Formulations: Operator-Entropy (285,733 at K=25; its explicit $D\times D$ Koopman matrix replaces the MLP predictor, -1.15% capacity), Reynolds-Stress (295,953, $+2.4\%$ for the covariance closure head), and Potential-Flow (338,117, $+17.0\%$ for the potential network). These specifications confirm performance differences do not stem from parameter scaling.
Linear AR Forecaster (Non-Neural Baseline): Ridge-regularized linear map from flattened context $\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{C\cdot K}$ to target windows $\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{S\cdot K}$, scored by observation-space prediction error under identical held-out EVT — isolating whether latent representation learning is necessary at all.

C. Multi-Tier Statistical Evaluation Metrics

Critical surveys [32], [39], [40] document severe reporting vulnerabilities in this literature — point-adjustment inflation, channel-averaging imbalances, threshold gaming — so we report metrics across complementary tiers:
Dataset-Macro Point-F1 (Primary Metric): Unweighted mean of Point-F1 across datasets:

$$\text{Macro-F1}_{\text{DS}} = \frac{1}{|\mathcal{D}|}\sum_{d\in\mathcal{D}}\left(\frac{1}{|C_d|}\sum_{c\in C_d}\text{F1}_c\right) \tag{16}$$

where C_d is the channel set of dataset d; this prevents high-stream datasets (GHL, 14 streams) from dominating and weights each physical domain equally.
Pooled Micro Point-F1: Global pooling of TP/FP/FN over all test timesteps, Micro-F1 $= 2\sum_c \text{TP}_c/(2\sum_c \text{TP}_c + \sum_c \text{FP}_c + \sum_c \text{FN}_c)$; it exposes models that flood long streams with false alarms. **Channel-Macro Point-F1** is the standard per-channel average. **PR-AUC** (average precision over all score thresholds) evaluates raw ranking independent of calibration. **Event Recall** is the fraction of contiguous anomaly segments $[t_{\text{start}}, t_{\text{end}}]$ in which at least one alarm fires. **Mean Detection Delay** Δt is the onset-to-first-alarm lag averaged over detected events (undetected events enter via Event Recall, not an arbitrary penalty). **Trivial all-positive reference**: a constant alarm scores $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at base rate π; models below it provide less value than an uncalibrated alarm. **Contiguous PA-F1**: point adjustment within contiguous ground-truth segments only; we also report test-sweep PA-F1 (best post-hoc threshold) to quantify the residual calibration gap.

D. Evaluation Regimes: Zero-Lookahead Causal vs. Buffered Inference

To keep early-warning capability separate from future-observation access, we evaluate two regimes: **strictly causal online detection** (at decision time T only $\{X_\tau\}_{\tau\le T}$ is accessible; scores use trailing attribution; zero lookahead) and **buffered window detection** (S=64 future timesteps are buffered before scoring, capturing full trajectory evolution at a deliberate 64-step latency). Both record Mean Detection Delay, Event Recall within delay budgets ($\Delta t \le 10, 64$), and false alarms per 1,000 steps ($\text{FAR}_{1\text{k}}$).

VI. Ablation Studies and Analysis

To isolate the individual contributions of each mathematical and architectural component, we conduct controlled ablation experiments across multiple telemetry regimes.

A. Ablation 1: Physical Regularizer Controls on Fixed Architectures

To test whether the physics-inspired regularizers benefit detection, Table I(a) reports controlled ablations across all 19 non-GHL streams (principal models at five seeds, controls at three, under the held-out protocol), disabling each physical term on a fixed backbone:

1) **Operator-Entropy vs. $\lambda_{\text{ent}} = 0$:** Disabling the spectral regularizer leaves Point-F1 unchanged (0.241 ± 0.014 vs. 0.244 ± 0.024; difference -0.003, 95% interval $[-0.011, +0.005]$, Wilcoxon $p = 0.62$); PR-AUC stays 0.347–0.348.
2) **Reynolds-Stress vs. $\alpha_{\text{stress}} = 0$:** Disabling the stress loss yields 0.203 ± 0.059 vs. 0.208 ± 0.071 ($+0.005$, $[-0.006, +0.016]$, $p = 0.41$). The null is also insensitive to the covariance estimator: per-sample temporal (`Reynolds-tempcov`) or batch (`Reynolds-batchcov`) covariances in place of the rank-1 outer product shift Point-F1 by $-0.009/-0.010$ and FPR by $+0.004/-0.042$ over nine streams — the second-moment term carries no signal beyond ordinary latent prediction.
3) **Potential-Flow vs. no-regime (K_r=1):** Disabling regime routing *improves* Point-F1 to 0.129 ± 0.061 from 0.096 ± 0.026 (-0.033, $[-0.060, -0.010]$, $p = 0.048$); both underperform TS-JEPA (0.234 ± 0.030, $p = 0.013$).

Across the full 19-stream suite, the physical regularizers produce no statistically detectable improvement over unconstrained latent prediction.

B. Ablation 2: Auxiliary Input-Channel Gating Variant

The auxiliary causal input-channel gate is compared against uniform pooling, a non-causal target-only leaky control, and independent context/target gating under identical TS-JEPA configuration (Supplementary Tables **??**, **??**; full regime definitions and per-fault-type results in Supplementary Section **??**). The effect is stream-dependent (MSL $0.047 \to 0.227$, SMAP $0.280 \to 0.019$); macro Point-F1 favors uniform weighting (0.118 vs. 0.077), and the leaky control's small margin ($+8.6\%$) shows the effect stems from quiescent-noise suppression rather than target statistics. The fault-type study finds gating roughly doubles point recall on transient faults at $\sim 3\times$ the nominal FPR, while quiet faults (flatline, stuck sensor, dropout) remain weakly detected under *every* regime (recall 0.02–0.07). Input gating is therefore reported as an evaluated variant; the deployed pipeline retains uniform input weighting.

C. Ablation 3: Littlewood-Paley Dyadic Wavelet Shells

Monolithic convolutions either over-smooth transients or miss long-period cycles (e.g., GECCO's diurnal consumption alongside abrupt pressure drops). As reported in Table I(b), replacing the standard front-end with the Fourier-domain dyadic shells ($\mathbf{x}_j = \mathrm{irfft}(\Psi_j \odot \mathrm{rfft}(\mathbf{x}))$) plus per-shell causal convolutions and learned routing yields a modest, stream-dependent effect under identical TS-JEPA configuration: macro buffered Point-F1 improves from 0.088 to 0.131 ($+49.6\%$ relative) across five multivariate streams, concentrated on MSL ($0.002 \rightarrow 0.052$) and SMD ($0.000 \rightarrow 0.177$), with a small SMAP regression ($0.294 \rightarrow 0.261$) and near-zero GECCO detection either way. Two caveats: the block adds $\approx 1.6 \times 10^5$ parameters (part of the gain may be capacity), and measured shell overlap (Section IV) means octave isolation is not attained. The front-end is a modest contributor, not the primary driver.

D. Ablation 4: Threshold Calibration Rules on Nominal Splits

The complete calibrator comparison across all 33 streams and all learned architectures is reported in Section VII-D and Table IV; we summarize the two findings most relevant to the ablation suite here. First, simple non-parametric rules match parametric SPOT: on TS-JEPA, a validation-maximum rule attains held-out FPR 0.0553 versus 0.0526 for standard SPOT at nearly identical Point-F1. Second, the kurtosis-adaptive multiplier degrades calibration, inflating TS-JEPA held-out FPR from 0.0526 to 0.1768 while leaving Point-F1 statistically indistinguishable (0.1694 vs. 0.1577). Section VII-D decomposes this failure into extremal clustering, chronological drift, and GPD goodness-of-fit breakdown.

E. Controlled Backbone Comparison: Latent Space vs. Observation Space

A foundational hypothesis is that predicting in a contracted latent space produces residuals better conditioned for EVT thresholding than observation-space reconstruction or forecasting. Comparing full JEPA models against public baselines confounds receptive field, depth, and normalization; Supplementary Table **??** therefore evaluates three formulations under a unified `HybridTCNEncoder` (C=256, S=64, d=32, 3-layer dilated causal TCN, LayerNorm, GELU, temporal pooling): **TCN-Obs-Recon** (reconstruction MSE), **TCN-Obs-Pred** (observation forecasting residual), and **TS-JEPA** (latent prediction against an EMA target), trained identically and calibrated via kurtosis-adaptive EVT on a held-out 20% nominal split.

Observation Reconstruction vs. Observation Forecasting: The matched-backbone results resolve whether false-alarm reduction stems from forecasting or from latent representation: TCN-Obs-Recon and TCN-Obs-Pred exhibit virtually identical tails (γ_2: 6.09/6.00 val, 57.26/55.66 test; clean FPR 27.29%/27.57%) and are statistically indistinguishable across the 15 streams (Wilcoxon clean FPR $p = 0.6784$; Point-F1 $p = 1.0000$). Forecasting in raw observation space does not condition residual tails because residuals directly reflect sensor jitter.

Latent Space Projection and Tail Compression: Latent-space prediction fundamentally reshapes the residual distribution: test excess kurtosis drops from 55.66 to 1.77 ($> 31\times$), and clean empirical FPR falls from 27.57% to 11.80% (Wilcoxon $p = 0.0479$, $\approx 6{,}500$ fewer false alarms per stream). Under 0-dB noise injection, observation-space FPR surges past 50% while TS-JEPA holds at 13.93% ($p = 0.0026$).

The Operational Sensitivity Trade-Off: This stability costs detection sensitivity: clean Point-F1 declines from 0.1916 (TCN-Obs-Pred) to 0.0785 ($p = 0.0054$), with recall dropping $0.4315 \rightarrow 0.1460$ — localized point anomalies that do not perturb macroscopic state are filtered with the noise. Latent prediction thus acts as a conservative state detector.

Investigation of Native Window Horizons: A secondary hypothesis is that baselines suffered under EVT calibration only because of the long W=320 window. Supplementary Table **??** refutes this: native horizons *worsen* tail conditioning. TranAD at W=10 spikes to $\gamma_2 = 448.25$ (vs. 29.93 at W=320) — micro-windows lack the context to average out sensor noise — and TimesNet at W=96 shows $\gamma_2 = 98.14$ with 44.09% empirical FPR. TS-JEPA retains well-conditioned residuals ($\gamma_2 = 3.14$, FPR 22.57%): observation-space tail corruption is inherent to raw reconstruction, not a window-length artifact.

VII. Results and Discussion

We report evaluations across 33 matched telemetry streams under the held-out calibration protocol: whether latent prediction conditions residuals for automated thresholding, the empirical behavior of the physics-inspired regularizers, operational latency, and the reliability of EVT calibration on real telemetry.

A. Benchmark Performance on Continuous Telemetry

Table II presents the empirical comparison across 33 continuous telemetry streams evaluated under the held-out calibration protocol of Section V-A (SPOT, $u = 0.98, q = 10^{-3}$).

Several primary empirical findings emerge from Table II:

False Alarm Suppression on Held-Out Telemetry: On the 19 non-GHL streams, latent predictors achieve held-out nominal FPR 0.076–0.086 versus 0.300 (TimesNet) and 0.307 (TranAD). Across all 33 streams (seeds averaged), the paired FPR reduction is statistically significant: TS-JEPA -0.153 vs. TimesNet (95% CI $[-0.247, -0.071]$, Wilcoxon p=0.002) and -0.151 vs. TranAD ($[-0.247, -0.070]$, p=0.003); Operator-Entropy shows -0.144 vs. TimesNet (p=0.006) and -0.142 vs. TranAD (p=0.006). Latent residuals avoid observation-level noise, yielding better-conditioned score distributions (isolated under the same-backbone control, Section VI-E).

Point-F1 and Continuous Ranking (PR-AUC) Parity: False-alarm suppression does not translate into statistically distinguishable detection gains. On the non-GHL subset TS-JEPA attains 0.234 Point-F1 / 0.323 PR-AUC versus 0.197/0.329 (TimesNet) and 0.130/0.287 (TranAD). Across all 33 streams the paired differences are inconclusive: vs. TimesNet $+0.012$ F1 (CI $[-0.053, +0.075]$, p=0.54), -0.000 PR-AUC (p=0.39); vs. TranAD $+0.051$ F1 (p=0.88), $+0.023$ PR-AUC (p=0.16).

Event Recall and Detection Conservatism: Lower FPR accompanies lower event recall: on non-GHL streams TimesNet captures 58.6% of events and TranAD 48.1%, versus 49.8%

TABLE I

ABLATION STUDIES. (A) MATCHED-CAPACITY REGULARIZER CONTROLS ON FIXED BACKBONES, 19 NON-GHL STREAMS (POINT-F1 = STREAM-MACRO MEAN ± SEED SD; HELD-OUT FPR ON UNSEEN NOMINAL DATA). (B) DYADIC-SHELL FRONT-END ON/OFF UNDER IDENTICAL TS-JEPA CONFIGURATION. AUXILIARY INPUT-CHANNEL GATING AND FAULT-TYPE RECALL PANELS ARE IN SUPPLEMENTARY SECTION **??**.

(a) Regularizer Controls on Fixed Architectures

Model / Control	Point-F1 (buffered)	PR-AUC	Held-out FPR	Event Recall
TS-JEPA (Unconstrained)	0.234 ± 0.030	0.323	0.076	0.481
Operator-Entropy JEPA	0.237 ± 0.013	0.323	0.086	0.506
$\lambda_{\text{ent}} = 0$ control	0.244 ± 0.024	0.347	0.083	0.518
Reynolds-Stress JEPA	0.207 ± 0.052	0.307	0.078	0.434
$\alpha_{\text{stress}} = 0$ control	0.203 ± 0.059	0.311	0.066	0.414
Potential-Flow JEPA	0.089 ± 0.036	0.201	0.091	0.303
no-regime control ($K_r = 1$)	0.129 ± 0.061	0.265	0.081	0.392
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490

(b) Dyadic-Shell Front-End On/Off

Stream	Standard Conv	Dyadic Shells	Δ
SMAP `P-3`	**0.294**	0.261	−0.033
MSL `M-1`	0.002	**0.052**	+0.050
SMD `machine-1-2`	0.000	**0.177**	+0.177
Daphnet `S01R01E1`	0.144	**0.165**	+0.021
GECCO `water_quality`	0.000	0.001	+0.001
Macro Mean	0.088	**0.131**	+49.6% rel.

TABLE II

HELD-OUT CALIBRATION BENCHMARK, 33 TELEMETRY STREAMS (7 DOMAINS). THRESHOLDS: SPOT (q=10^{-3}) ON HELD-OUT NOMINAL SPLITS. METRICS: HELD-OUT NOMINAL FPR (ON DATA THE THRESHOLD NEVER SAW), UNADJUSTED POINT-F1, PR-AUC, EVENT RECALL, MEAN DETECTION DELAY Δt. "NON-GHL": STREAM-MACRO OVER 19 STREAMS AT FIVE SEEDS; "GHL": 14 STREAMS AT THREE-TO-FIVE SEEDS; "DATASET-MACRO": UNWEIGHTED MEAN OF THE 7 DATASET MEANS. BEST IN **BOLD**, SECOND UNDERLINED.

	Non-GHL Streams (19 Streams, 5 Seeds)					Summary	Gasoil Heating Loop (14 Streams)	
Model Architecture	**Held-Out FPR ↓**	**Point-F1 ↑**	**PR-AUC ↑**	**Event Recall ↑**	**Mean Delay (Δt) ↓**	**DS-Macro F1 ↑**	**Held-Out FPR ↓**	**Point-F1 ↑**
Latent-Space Predictive Architectures (JEPAs)								
TS-JEPA (Unconstrained Control)	0.076	0.234	0.323	0.481	80.3	**0.148**	0.005	0.000
Operator-Entropy JEPA	0.086	**0.237**	0.323	0.506	192.3	0.130	0.012	0.001
Reynolds-Stress JEPA	0.078	0.207	0.307	0.434	132.8	0.122	**0.003**	0.002
Potential-Flow JEPA	0.091	0.089	0.201	0.303	146.9	0.065	0.011	0.006
Observation-Space Reconstructive Baselines								
TimesNet (ICLR 2023)	0.300	0.197	**0.329**	**0.583**	48.5	0.100	0.061	**0.022**
TranAD (VLDB 2022)	0.307	0.130	0.287	0.490	234.4	0.069	0.046	0.020
Non-Neural Linear Baseline								
Linear AR Forecaster	**0.006**	0.057	0.262	0.478	**0.0**	0.038	**0.002**	0.000

(TS-JEPA), 39.8% (Reynolds), 33.1% (Potential-Flow). Subtle onsets that do not shift dominant latent coordinates are filtered with the noise.

Detection Delay Trade-Offs (Δt): Mean detection delay (Table II) varies widely: TimesNet responds fastest (Δt=48 steps), TS-JEPA averages 80, and the physics variants are slower still (133–192); TranAD is slowest (234). Because latent encoders smooth observation noise, alarms trigger only once anomalies propagate into dominant latent coordinates — deployments trade 4× fewer nominal false alarms for added initial latency.

Necessity of Nonlinear Representation Learning (Linear AR Control): A ridge-regularized linear AR forecaster under identical calibration attains FPR 0.006 but collapses to Point-F1 0.057 (precision 4.35%): observation-space linear dynamics cannot separate faults from background variance. Nonlinear latent representation is necessary to combine controlled false alarms with usable detection.

Domain-Specific Telemetry Dynamics: Supplementary Tables **??**–**??** report the per-dataset Point-F1/PR-AUC breakdown. Relative performance depends on the physical sensing regime: TimesNet leads on Daphnet periodic kinematics (0.245 F1, 0.343 PR-AUC vs. TS-JEPA 0.155/0.137), where 2D Fourier decomposition captures multi-frequency gait structure; latent prediction leads on spacecraft telemetry (SMAP: TS-JEPA 0.441 F1 / OpEntropy 0.591 PR-AUC; MSL: OpEntropy 0.281 F1 / 0.445 PR-AUC, versus TimesNet 0.304/0.159 F1). On SWAN, TimesNet triggers zero test detections under severe threshold miscalibration (0.000 F1 despite 0.292 PR-AUC). On GHL and Metro, all models struggle (0.000–0.022 F1): subtle actuator drifts against stationary backgrounds and extreme traffic surges evade static EVT bounds respectively.

B. Evaluating Physical Inductive Priors: Hypothesis Testing and Null Results

Does adding physics-inspired regularizers (Koopman spectral entropy, Reynolds-stress closure, curl-free potential flow) improve detection over unconstrained `TS-JEPA`? The 33-stream data refute this:

1) **Operator-Entropy vs. TS-JEPA:** paired difference +0.002 Point-F1 (CI $[-0.025, +0.029]$, p=0.71), +0.002 PR-AUC (p=0.86), +0.009 FPR (p=0.75); leads on only 9 of 33 streams.
2) **Reynolds-Stress vs. TS-JEPA:** −0.015 Point-F1 ($[-0.042, +0.007]$, p=0.61), −0.007 PR-AUC (p=0.13), +0.001 FPR (p=0.92); leads on 11 of 33.
3) **Potential-Flow vs. TS-JEPA:** significantly worse Point-F1 (-0.081, $[-0.160, -0.018]$, p=0.033) and nominally lower PR-AUC (-0.067, p=0.58); leads on 7 of 33.

These findings align with the matched-capacity regularizer controls in Table I(a) and Section VI-A, where disabling each physical loss term on a fixed backbone produces changes well within seed noise. We conclude that the physical regularizers provide no measurable anomaly detection advantage over unconstrained latent prediction. Performance differences between models stem from shared representation conditioning rather than the physical loss terms.

The dynamical diagnostics evaluating *why* each regularizer fails (operator rank preservation without detection gain, covariance-shift insensitivity, and curl-free misspecification on

cyclic telemetry) are reported in Supplementary Section **??** (Tables **??**–**??**).

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Event-level (incident) metrics, operational latency under dense vs. buffered inference, and rolling-SPOT adaptation case studies on Room-Occupancy, SMAP `P-3`, CalIt2, and a five-stream causal-vs-buffered panel are reported in Supplementary Section **??** (Tables **??**, **??**): dynamic rolling SPOT suppresses false alarms up to $21\times$ on drift-prone streams at the cost of event recall (100% $\rightarrow$ 63.6%), and TS-JEPA causal false-positive rate (4.18%) is an order of magnitude below reconstructive baselines (20.8%–30.1%).

Table III reports the complete 33-stream benchmark under strictly causal (0-lookahead) streaming. On the 19 non-GHL streams, unconstrained TS-JEPA reduces held-out nominal FPR to 0.0136 (versus 0.2427 for TimesNet and 0.2219 for TranAD), an out-of-sample false-alarm suppression of roughly 18-fold; on the 14 GHL streams causal FPR falls below 0.009 for JEPAs and 0.04–0.06 for baselines, but causal Point-F1 there is near-zero for latent predictors (≤ 0.006 vs. 0.022 TimesNet). Causal Point-F1 drops to 0.1118 ± 0.0412 for TS-JEPA and 0.1093 ± 0.0085 for Operator-Entropy, whereas TimesNet achieves 0.2091 ± 0.0265. Event recall likewise declines from 58.82% (TimesNet) to 46.55% (TS-JEPA). This confirms that without a prospective lookahead buffer to observe multi-step trajectory evolution, latent predictive scoring functions as an ultra-conservative filter that prevents false alarms at the price of reduced detection sensitivity.

a) Buffered Early-Warning Lead Time: To separate prospective warning capability from detection fidelity, we measure the fraction of events first alerted before their labeled onset under buffered inference across five streams. TS-JEPA warns 12.3% of events pre-onset (7 of 57 pooled), with a median lead time of 180 steps and 23.8 false warnings per 1,000 nominal steps. TimesNet warns 73.7% of events pre-onset at a comparable median lead (204 steps) but emits 206.4 false warnings per 1,000 nominal steps; on GECCO it alerts 37 of 37 events early while generating 873.6 false warnings per 1,000 nominal steps, approaching a constant-alarm regime. Pre-onset coverage is therefore confounded by the underlying alarm rate: at its conservative operating point, TS-JEPA trades early-warning coverage for an order-of-magnitude lower nuisance-alarm rate.

D. Automated Threshold Calibration Realities on Non-Stationary Telemetry

A critical requirement for autonomous telemetry monitoring is whether automated extreme-value thresholding adheres to theoretical design risk parameters. When practitioners specify a design risk of $q = 10^{-3}$ (0.1%), the operational expectation is that nominal false alarms will occur at or below this frequency.

Table IV compares threshold calibrators on identical score streams across all 33 streams under buffered inference. Four realities emerge:

1) **Systematic design-risk overshoot:** SPOT (u=$0.98, q$=10^{-3}) never achieves its design risk out-of-sample — held-out nominal FPR averages 5.26% (TS-JEPA), 6.17% (OpEntropy), 22.17% (TimesNet), 22.17% (TranAD), a 53–$222\times$ overshoot.
2) **High failure rate:** 95% bootstrap CIs exclude q in 61%/65%/86%/82% of runs (TS-JEPA/OpEntropy/TimesNet/TranAD); non-stationary drift violates EVT's stationary-tail assumptions.
3) **Parity with empirical quantiles:** on TS-JEPA, validation-maximum gives FPR 0.0553 / F1 0.1591 and empirical p99.5 gives 0.0682 / 0.1598 — parametric GPD tail modeling confers no demonstrable benefit.
4) **Kurtosis-adaptive failure:** the multiplier inflates TS-JEPA FPR from 0.0526 to 0.1768 (CI-exclusion $61\% \rightarrow 82\%$) with indistinguishable F1 (0.1694 vs. 0.1577); calibration-split kurtosis does not forecast out-of-sample tail shifts.

To separate the candidate causes of calibration failure, we recorded per-stream tail-fit diagnostics on a nine-stream non-GHL subset (TS-JEPA and TimesNet, three seeds each; `reports/evt_tail_diagnostics.csv`). Three observations disentangle the mechanism. First, tail-sample scarcity is genuine: the median exceedance count above $u = p_{98}$ is only 21 (range 16–70), and the Grimshaw estimator returns the degenerate boundary shape $\xi = -0.9$ in 33% of TS-JEPA and 52% of TimesNet fits, so tail-estimation error is real. Second, goodness-of-fit is nonetheless not the dominant failure: a Kolmogorov–Smirnov test of the excesses against each fitted GPD rejects at $p < 0.05$ in only $6/27$ (TS-JEPA) and $8/27$ (TimesNet) cells. Third, realized held-out FPR varies from 0.0 to 0.67 across seeds on the *same* stream under fits that pass the KS test — direct evidence that inter-block distribution drift, not tail mis-specification, drives the overshoot. Exceedances are also strongly clustered (mean of 10–22 exceedances per cluster, more pronounced for TimesNet, whose residuals spike in sustained bursts), which inflates the effective variance of the FPR estimate beyond what an i.i.d. tail model assumes. Finally, the empirical 99.5th-percentile rule achieves an equal or lower held-out FPR than SPOT in roughly half the cells ($13/27$ TS-JEPA, $16/27$ TimesNet), confirming that parametric tail extrapolation confers no reliability advantage under this drift regime.

a) Extended calibration diagnostics: The multi-parameter EVT sensitivity sweep (u, q, kurtosis coefficient β) and the four-mechanism failure decomposition (extremal clustering via the Ferro–Segers extremal index, two-sample KS drift, GPD goodness-of-fit rejection, and empirical risk overshoot) are reported in Supplementary Section **??** (Tables **??** and **??**). The decomposition attributes the design-risk overshoot primarily to inter-block distribution drift and extremal clustering rather than tail mis-estimation, consistent with the per-stream tail diagnostics above.

Computational profiling (parameters, VRAM, inference latency, throughput) is reported in Supplementary Table **??**: the JEPA family uses 183.8–185.3 MB peak memory (nearly $7\times$ less than TranAD) at 0.60–0.84 ms per-window inference.

TABLE III

STRICTLY CAUSAL STREAMING BENCHMARK (0-LOOKAHEAD, TRAILING ATTRIBUTION), 33 STREAMS, HELD-OUT SPOT (q=10^{-3}). "NON-GHL": 19 STREAMS AT FIVE SEEDS (STREAM-MACRO, $\pm$ SEED SD); "GHL": 14 STREAMS AT THREE-TO-FIVE SEEDS. LATENT PREDICTORS CUT NOMINAL FALSE ALARMS TO 0.97%–1.89% ON NON-GHL (12–25$\times$ BELOW RECONSTRUCTIVE BASELINES) AT THE COST OF LOWER CAUSAL POINT-F1 AND EVENT RECALL.

	Non-GHL Streams (19 Streams, 5 Seeds)					Gasoil Heating Loop (14 Streams)		
Model Architecture	**Held-Out FPR ↓**	**Point-F1 (Mean ± SD) ↑**	**PR-AUC ↑**	**Event Recall ↑**	**Test FPR ↓**	**Held-Out FPR ↓**	**Point-F1 ↑**	**Event Recall ↑**
Latent-Space Predictive Architectures (JEPAs)								
TS-JEPA (Unconstrained Control)	0.0136	0.1118 ± 0.0412	0.2672	0.4655	0.0844	0.0035	0.000	0.000
Operator-Entropy JEPA	0.0189	0.1093 ± 0.0085	0.2528	0.4692	0.0986	0.0085	0.001	0.024
Reynolds-Stress JEPA	0.0111	0.0876 ± 0.0367	0.2700	0.4563	0.0842	**0.0023**	0.002	0.014
Potential-Flow JEPA	**0.0097**	0.0233 ± 0.0166	0.1775	0.3428	**0.0289**	0.0047	0.006	0.032
Observation-Space Reconstructive Baselines								
TimesNet (ICLR 2023)	0.2427	**0.2091 ± 0.0265**	**0.2931**	**0.5882**	0.2163	0.0563	**0.022**	**0.850**
TranAD (VLDB 2022)	0.2219	0.1405 ± 0.0154	0.2483	0.5334	0.2027	0.0398	0.018	0.811

TABLE IV

CALIBRATOR COMPARISON ON IDENTICAL SCORE STREAMS (33 STREAMS, BUFFERED INFERENCE, UP TO FIVE SEEDS). DESIGN RISK q=10^{-3}=0.1%; THE CI-EXCLUSION COLUMN SHOWS AUTOMATED CALIBRATION SYSTEMATICALLY OVERSHOOTS OUT-OF-SAMPLE.

Model Architecture	**Calibrator Rule**	**Held-Out Nominal FPR**	**CI Excludes $q = 10^{-3}$**	**Point-F1**
TS-JEPA (Control)	Standard SPOT ($q = 10^{-3}$)	**0.0526**	61%	0.1577
	SPOT + Kurtosis ($\beta = 0.167$)	0.1768	82%	**0.1694**
	Empirical 99.5th Percentile	0.0682	55%	0.1598
	Validation Maximum	0.0553	65%	0.1591
Operator-Entropy	Standard SPOT ($q = 10^{-3}$)	**0.0617**	65%	0.1599
	SPOT + Kurtosis ($\beta = 0.167$)	0.1229	76%	0.1441
	Empirical 99.5th Percentile	0.0758	62%	**0.1658**
	Validation Maximum	0.0656	64%	0.1627
TimesNet	Standard SPOT ($q = 10^{-3}$)	**0.2217**	86%	0.1395
	SPOT + Kurtosis ($\beta = 0.167$)	0.2281	82%	0.1352
	Empirical 99.5th Percentile	0.2338	87%	**0.1413**
	Validation Maximum	0.2263	86%	0.1410
TranAD	Standard SPOT ($q = 10^{-3}$)	**0.2217**	82%	0.0936
	SPOT + Kurtosis ($\beta = 0.167$)	0.2453	84%	0.0940
	Empirical 99.5th Percentile	0.2343	84%	**0.0962**
	Validation Maximum	0.2270	84%	0.0952

VIII. CONCLUSION AND FUTURE WORK

We investigated empirically whether JEPA-style latent prediction and physics-inspired regularizers resolve the false-alarm vulnerability of multivariate anomaly detection, evaluating 33 telemetry streams across seven physical domains under a strictly held-out calibration protocol. Three conclusions follow.

First, prediction in a contracted latent space conditions residual distributions: filtering observation-space jitter cuts held-out nominal FPR roughly four-fold versus reconstructive baselines (-0.153 vs. TimesNet, p=0.002; -0.151 vs. TranAD, p=0.003, five-seed subset), while Point-F1 and PR-AUC remain statistically indistinguishable ($p \geq 0.13$). Under strictly causal streaming, latent prediction suppresses false alarms to 1.0%–1.9% at the cost of lower F1 and delayed event recall.

Second, hypothesis testing refutes a detection advantage for the physics-inspired regularizers: neither Operator-Entropy nor Reynolds-Stress outperforms TS-JEPA ($p \geq 0.61$) across all 33 streams, and Potential-Flow is significantly worse (p=0.033), unable to represent circulating orbits. Matched-capacity controls confirm that disabling each physical term shifts Point-F1 by only $+0.001$ to $+0.006$ ($p \geq 0.58$); the formulations enforce their intended mathematics but real telemetry does not obey the idealized laws.

Third, SPOT calibration systematically overshoots its design risk on non-stationary telemetry (5.3–6.2% nominal FPR for JEPAs, $\approx$ 22% for baselines at q=10^{-3}; CIs exclude q in 61–86% of runs). Parametric tail modeling offers no advantage over empirical quantiles, and kurtosis adjustment degrades rather than corrects the drift.

Scope. All evaluations are *within-dataset*: models are trained, calibrated, and tested on disjoint chronological partitions of the same stream. We claim nothing about cross-dataset transfer or unseen fault types (cf. Section **??**). Extensions include solenoidal components via Helmholtz decomposition, memory-bounded online drift adaptation, accelerated streaming inference kernels, and integer quantization for edge deployment.

CODE AND DATA AVAILABILITY

All benchmark extraction scripts, preprocessing pipelines, checkpoints, training implementations, evaluation runners (`scripts/run_multiseed_heldout_benchmark.py`, `scripts/run_protocol_ablation.py`), diagnostic artifacts, and logs are released at https://anonymous.4open.science/r/NCAD_CS-Reproducibility-8F21; public datasets are accessible via their referenced repositories.

REFERENCES

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[4] M. P. Cuéllar, M. Santos, J. L. Alonso, M. Fabregas, and R. Farías, "Explainable anomaly detection in spacecraft telemetry," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108119, 2024.

[5] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[6] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[7] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.

[8] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[9] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.

[10] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[11] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[12] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.

[13] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.

[14] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.

[15] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[16] D. Salinas, V. Flunkert, J. Gasthaus, and T. Januschowski, "DeepAR: Probabilistic forecasting with autoregressive recurrent networks," *International Journal of Forecasting*, vol. 36, no. 3, pp. 1181–1191, 2020.

[17] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[18] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.

[19] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.

[20] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.

[21] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.

[22] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[23] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.

[24] E. Eldele, M. Ragab, Z. Chen, M. Wu, C. K. Kwoh, X. Li, and C. Guan, "Time-series representation learning via temporal and contextual contrasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 2352–2359.

[25] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Wang, L. Liu, and L. Sun, "Self-supervised time series representation learning: A systematic survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3241–3261, 2024.

[26] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[27] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.

[28] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.

[29] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[30] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

[31] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.

[32] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.

[33] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.

[34] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.

[35] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[36] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.

[37] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

[38] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.

[39] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.

[40] R. Wu and E. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2434, 2023.